



Evaluating Reports

Many reports use data sets to support a point of view. If you are able to evaluate the data with a critical eye, you can determine how valid this point of view is.

The Reason for the Study

To critically judge a report that includes data, you must look at many factors. First determine who carried out the study and for what reasons.

EXAMPLE 1

The Olive Oil Society of the Americas reports that countries in which people consume olive oil daily have a lower rate of heart disease. Why did the organization release the report?

SOLUTION

Think about the name of the organization that produced the report. The Olive Oil Society of the Americas most likely wishes to promote the use of olive oil. The organization can achieve that goal if consumers believe that olive oil has tangible health benefits. ■

How Were the Data Collected?

A sample is **biased** if it is not representative of its population. For instance, a farmer wants to know the depth of the roots in a certain crop of plants. Because they are nearest to his house, he digs up 6 plants along the southwest corner of the field and inspects their roots.

► **Remember** We choose to study a sample only when we can't study the entire population.

The sample is likely biased because all the plants come from the same area of the field. This area may receive more or less sunlight than other areas. The amount of sunlight a plant receives affects the plant's growth and therefore its root system. Soil conditions may also differ in this part of the field. Also, 6 is a small number for a sample.

Bias may also arise from the question or questioning technique used. For example, a question may give only one side of the story. Compare these questions:

- **Biased toward a No response** Pesticides cannot be washed off of all foods, and pesticides leak into nearby waterways. Do you think our farmers should be allowed to put our health at risk by using pesticides?
- **Biased toward a Yes response** If our farmers cannot control pests effectively, our local food production could drop by 35%, creating a food shortage and economic recession. Do you agree that our farmers need to use pesticides?

Bias can also arise from questioning that is too personal, is too long and wordy, or has only a limited number of responses (because of multiple choice).

The Relevancy of the Data

Even a study that is well designed and carried out will have limitations on what can be learned from the study.

EXAMPLE 2

The Olive Oil Society of the Americas reports that countries in which people consume olive oil daily have a lower rate of heart disease. What can be concluded from this study?

SOLUTION

There is a difference between correlation and causation. Although it may be true that heart disease is lower in countries where olive oil is consumed daily, it does not necessarily mean that consuming olive oil causes this reduction in disease. You can only conclude that there may be a link between daily use of olive oil and a reduced risk of heart disease.

The olive oil report could be a stepping stone for further study. Which countries have lower rates of heart disease? Do the countries share a certain type of climate? Are there other dietary factors to consider? What about exercise or other health factors? Is there a scientific reason why the chemicals in olive oil would positively affect the human heart?

Statistics and data are useful for helping you understand many situations, but you need to be aware of how those data are collected and presented to gain a deeper understanding of reports. ■

► **Think About It** There are many factors that affect health and heart disease. Diet is just one of them.